



# RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

## Department of Information Technology Academic Year 2025 - 2026 (Odd Semester)

**Degree, Semester & Branch:** III Semester B.Tech. IT

**Course Code & Title:** CD3291 & Data Structures and Algorithms

**Name of the Faculty member (s):** Mrs. G.Mareeswari, AP/IT

### Innovative Practice Description

- **Unit / Topic:** Unit II / Stack
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO6,8
- **Activity Chosen:** One Minute Paper

- **Justification:**

The stack is a fundamental linear data structure that follows the Last-In-First-Out (LIFO) principle, making it essential for understanding how data is managed in constrained access scenarios. It plays a crucial role in various core areas of computer science, such as function call management, expression evaluation, syntax parsing, backtracking algorithms, and memory management. Due to its simple operations—push, pop, peek, and isEmpty—stacks provide an excellent foundation for grasping more complex data structures and algorithmic thinking. Choosing the stack for this one-minute paper activity helps students solidify their understanding of abstract data types, develop their ability to trace program flow, and appreciate real-world applications. This concise yet powerful topic encourages students to reflect on how such a basic structure supports many critical system operations and algorithm designs.

- **Time Allotted for the Activity:** 10 Minutes

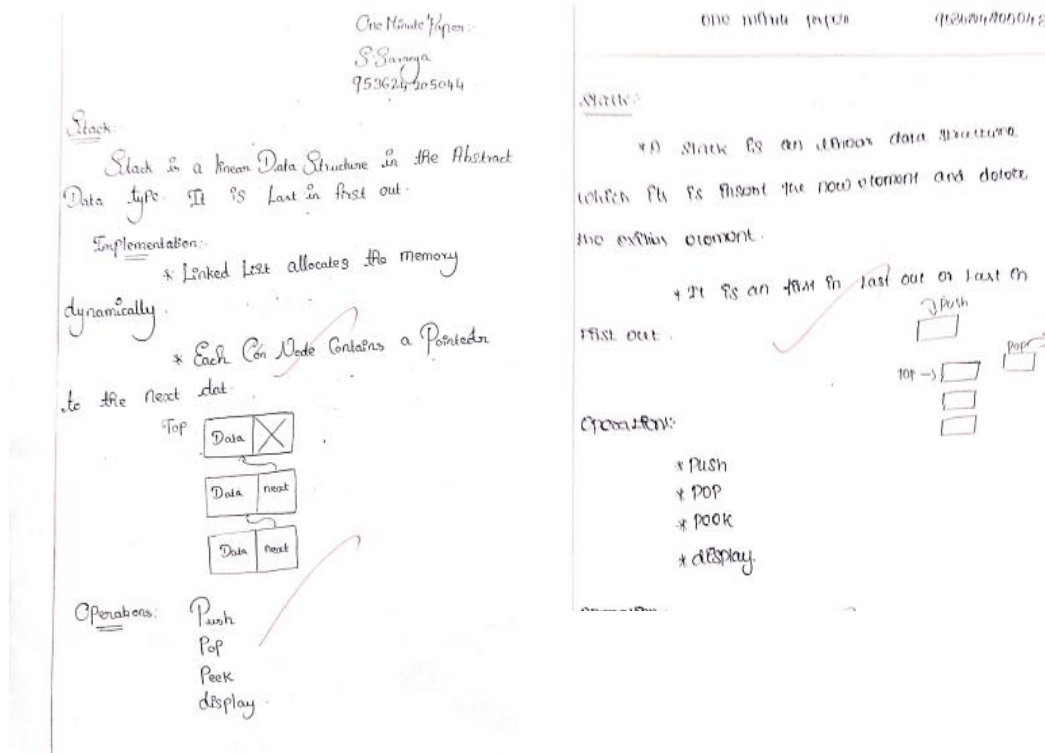
- **Details of the Implementation:**

. At the end of the class on the stack data structure, students took part in a one-minute paper activity to share what they learned. The instructor had explained the LIFO (Last-In-First-Out) rule, basic stack operations like push, pop, and peek, and how stacks are used in real life, such as in undo buttons, function calls, and checking brackets. Students were asked to write what they understood and informed to draw the basic structure of stack. Many students said they understood the main operations and how a stack works. This short activity helped students think about the topic and remember it

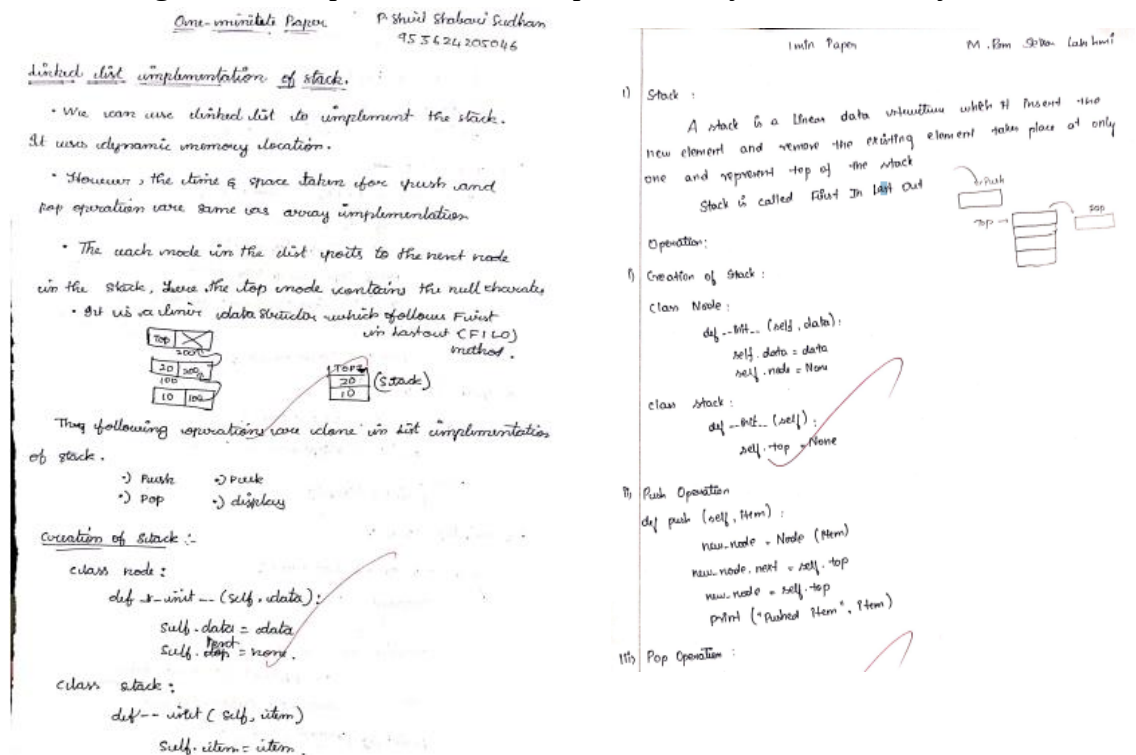
better. It also helped the teacher know what to explain more.

The points shared by students are shown in Figure 1a and Figure 1b.

- **Images / Screenshot of the practice:**



**Figure 1.a** Sample One-Minute Paper of Saranya. S and Santhiya. S



**Figure 1.b** Sample One-Minute Paper of Shree Shabari Suthan. P and Ram Selva Lakshmi. M

• **CO – PO / PSO mapping:**

| CO  | PO1 | PO2 | PO3 | PO9 | PSO1 |
|-----|-----|-----|-----|-----|------|
| CO4 | 3   | 1   | 1   | 1   | 1    |

(1 – Low      2 – Moderate      3 – High)

• **PO / PSO mapped:**

| Innovative practice                  | PO1                                                                      | PO2                                                                                                     | PO3                                                                                         | PO9                                                 | PSO1                                                                                                    |
|--------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| CO2                                  | 3                                                                        | 1                                                                                                       | 1                                                                                           | 1                                                   | 1                                                                                                       |
| <b>Justification for correlation</b> | Needs engineering and mathematical knowledge to understand Stack concept | Stack concept is used in analysis of solving complex engineering problems example expression conversion | The students can able to find solution for the complex problems using the concept of Stack. | The students individually involved in this activity | The students can apply the concept of Stack in solving various problems of data science and algorithms. |

**Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

- Students shared that the activity helped them think more deeply about the topic and express their understanding in a simple, clear way.
- After completing the activity, students reported feeling more comfortable and better able to remember the main ideas covered in the class.

❖ ***Benefit of the practice:***

- It helps students to understand and learn better.
- Teachers can quickly understand what students have understood and what they haven't.
- Students can find out what they know well and what they need to concentrate more.

❖ ***Challenges faced in implementation:***

- Some students had a good understanding of the stack concept but found it challenging to clearly express their thoughts in writing.
- Because of the limited time, a few students were unable to include the structure of the stack in their responses.

**References:**

- ❖ <https://www.ritrjpm.ac.in/images/B.Tech-IT/2024-2025/GM CD3291 One Minute Paper.pdf>
- ❖ <https://www.rochester.edu/college/cetl/faculty/one-minute-paper.html>
- ❖ <https://www.unl.edu/gradstudies/current/teaching/minute>
- ❖ [https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1 GM CS335 3\\_Functions\\_OMP.pdf](https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1 GM CS335 3_Functions_OMP.pdf)
- ❖ <https://oncourseworkshop.com/self-awareness/one-minute-paper/>

**Signature of Faculty Member**

**HOD**